Electricity and The Grid

- are we social creatures, or selfish? A primer for the grid.

I fear we may selfish, or perhaps just ignorant.

Yes, that's it — ignorant. Which results in our seeming selfish. Maybe I can help. Knowledge, not mine, Larry Blair's. https://substack.com/@newzealandenergy Larry knows heaps more than I, but this point above all is really important, and needs saying:

GET BATTERIES BEFORE YOU GET SOLAR,

NEVER GET SOLAR WITHOUT BATTERIES.

If you accept that, and will do it, you need read no further.

First the very simple explanation: The grid was made before rooftop solar. It was designed to make electricity at hydro works like Benmore and the later Meremere coal-fired generators and feed it into our homes and businesses. Trying the other way around makes problems. Why? Most Kiwi blokes can tow a trailer. With the car in charge it all goes well, but when the trailer pushes the car? We know better than to let that happen. When the household rooftops push the grid? Similar problems — it wasn't designed for it. If you are happy with that explanation, you need to read no further.

Why are batteries good? Simple: smooth, steady electricity use is easiest to provide. Common sense really. But the world isn't that simple, because our generators and grid are fully used only at dusk, when every house in the country starts cooking dinner. We have had to pay for generators and grid that are virtually idle all the rest of the night. How could we reduce this 'peak problem'? With batteries. How could that be done? Very basically, equip every house with a battery to run the home at dinner time and early evening, and charged it during the rest of the night, that would reduce the peak demand, reduce the capacity we need (and have to pay for) in our supply chain.

However, we would need the same amount of water through our hydro works each day - but more evenly used over the 24 hours.

Can rooftop solar help? Yes, but with some caveats. Every day that it's used to recharge the batteries, it reduces the water usage from our dams. But does rooftop solar reduce the size and capital cost of our grid and generators? Alas, no. And why not? Because on sunless days we still need night-recharging of our batteries, and we have to allow for days when the whole country is under cloud, and windless, too. Shame, but there is no way past it — unless we instal huge, multi-week batteries, country-wide, and huge solar and wind farms that can quickly recharge the huge batteries. And that takes us back again into high capital costs.

Now the worst case: nationwide rooftop solar without battery. Why so? Because at sunny noon there is no domestic demand from the grid, which still must maintain and synchronise supply. When the sun goes down there is still the evening peak, but the peak effect is worse, because the contrast with daytime usage level is much more severe. As well, in order to export power each of our rooftops must output above the street grid voltage, to drive our electricity outwards (so to speak). Which minutely raises the street voltage, so our neighbours' must export at a slightly higher voltage still. When that happens over a big area the voltages rise beyond the design limits, transformers would fail due to over-voltage, the safety switches are triggered and that part of the grid shuts down. When the whole country became so equipped, the historical "one-way' grid would become unmanageable.

However, the grid can be redesigned to be two-way, which is far from cheap — it uses masses of complex and expensive gear to do it. Which is going to be needed in every town, if significant numbers of us went 'solar-without-battery'. Me, I think that should be forbidden, but then, I don't mind being called anti-social, because I know that it's going solar without battery that is the real anti-social choice. You can go now.

Do we know enough about the way we actually power our homes and industries? I thought I did, but I had learnt about that, 70 years ago when I was thinking about studying engineering. It's simple in concept, hugely complicated in practice and getting more so with each passing year, with the increasing proliferation of batteries, with more wind farms and solar, both solar farms and solar rooftop.

The original basics of the system, crudely put, are that electricity is generated by changing the magnetism near a wire. When that is done by moving the wire and magnet past each other, we call that a generator. When that is done by using a water turbine to spin a huge magnet within a circular set of coils (the output coils) wound around a great iron cylinder, we call it a hydroelectric generator, and there are numbers of them below each of the dams that provide the water pressure, in each of the hydro works dotted around the country. Our coal-powered stations drive the generators with steam turbines instead of water turbines, but the principle is the same.

The resulting current alternates ("AC") because magnets have a North pole and South pole which drive the output current in alternating directions. This is fortunate, because it's AC transformers that make "the grid" feasible, by raising the generated voltage to up to 220 thousand volts, and so reducing the current by a factor of many thousands. Current, through a resistance, swaps electricity for heat. Reducing the current, by raising the voltage, reduces the losses caused by the resistance of a grid of thousands of miles of cables, strung between pylons, spanning the country. At the city end of the grid, progressive step-down transformers lower the voltage, the last transformer stage, down to 230 volts, being on the residential block, serving one cluster of homes. More on that later.

The generators are designed to produce AC at 50 cycles per second. That means that in one fiftieth of a second, the voltage goes from flowing one way, to the other, and back to the first. Every generator in the country, and all exactly in phase. All go from current one way to back and forth at precisely the same time, to within a few microseconds. They are kept in step by the alternating voltage, and their huge masses of rotating iron are huge flywheels linked by electricity instead of steel shafts.

Now we come to the excellent role of Transpower. This is all a bonus topic. Not required reading at all. I'm including it because Transpower is our unsung hero. If it were not for what Transpower does, we would have no effective electricity supply. Transpower looks after the design, building, operation, maintenance and protection of the grid.

The design of extension and modifications to the grid and its control and safety mechanisms, goes through stages of consideration, consulting, analysis and final approval. Being outside government, Transpower is not jerked around every three years. Although it is a State-Owned-Enterprise, it is run by an independent Board of Directors, with the Commerce Commision regulating its revenue and pricing, and the Electricity Authority controlling its market operations.

But, above all, Transpower keeps the grid humming, literally, at 50hz, proving 230volts at the household, for whatever the quantity being drawn - but it does change the prices as the day proceeds, to discourage excessive use and keep demand within controllable bounds.

The unseen magic that makes this possible is a huge clutch of clever electronic and heavy engineering systems that I will now list very briefly for you to research yourself, if you get the bug.

Keeping your clock accurate: Frequency Control: MFK: Multiple Frequency Keeping within SCADA Supervisory Control And Data Acquisition. Send signals to generators to adjust their output. FKC: Frequency Keeping Control within the HVDC High Voltage Direct Current Cooks Strait cable so the north and south island stay in step. PLSR: Partially Loaded Spinning Reserve uses partially loaded generators to provide more or less output for MFK to keep the clocks in time. IL: Interruptible Load, if the system grunts under heavy load and slows down,there are large factories that can be tripped off-line to allow the system to pickup speed again. AUFLS: Automatic Under-Frequency Load Shedding, which progressive drops load off the grid - better a few areas be without power for a short time, than the whole country shutting down!

Matching supply with demand: Load Balancing: SPD: Scheduling, Pricing and Dispatch mathematical model, every five minutes it recalculates using generation costs, reserves, demand and grid limits and losses, to determine the lowest cost solutions at those times. RTD: Real Time Despatch, every five minute it remotely alters generator output using ICCP Inter control Centre Communications Protocol. PRSS: Price-Responsive Schedule load forecasting based on historical patterns, revenue meters, current weather and the five-minute load analysis as an input to the computations. DSP: Demand Side Participation allow big users to bid for usage as the pricing varies, to further balance demand with the available supply.

Safety from lightning or overload: Transient Protection: SIPS: System Integrity Protection Schemes, deals with sudden circuit or transformer changes, within milliseconds trips generators or drops load to prevent overload or system collapse. PSS: Power System Stabilisers, modify generators' regulators to prevent 'hunting' resulting from the sudden load of a transient fault. AVR: Automatic Voltage Regulators continuously regulate the control of each generator's output voltage. Synchronising Auto Reclosers: when a lightening strike or other sudden fault (as a pilot my mind turns to an aircraft flying into the wires) causes a sudden shut down on a high pylon line, these units ensure the circuit-breaker will only be rejoining the ends when they are in synch. Dynamic Modelling, including Transient Rotor Angle Stability studies provide information from simulations, that are used as inputs for 'tuning' the protection methods (toward the hypothetical optimum - always being sought).

I haven't even mentioned the dependable old synchronous rotary condensers one of which I saw running at Islington perhaps 70 years ago and still in use at Haywards for the Cook Strait link and the massive Benmore generators which perform the same function. The massive rotating armatures acts as flywheels for the grid, smoothing the alternating waveform and holding the voltage steady.

Rooftop solar, batteries, windfarms and the ways they impose changes to the patterns and locations of generation and demand, these add considerably to the complexity of these already complex issues. They are already driving changes to the transmission lines, substation methods and control protocols.

It will get worse.